

GEOPHYSICAL FLOWS



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Book: Geophysical Flows (Omta)

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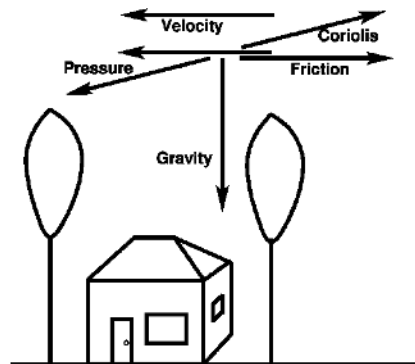
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1: General concept- Newton's 2nd law

The wind blowing over your house, the Gulf Stream flowing through the North Atlantic Ocean from Florida to Europe, the jet stream at 10 km altitude... All these different flows in the atmosphere and in the ocean can be understood using ideas from the physics of fluids.



A fluid is any substance that is able to flow which can be a gas, a liquid, or in some cases even a solid. In reality, fluids consist of molecules, particles that interact with each other. However, fluid mechanics deals with very large collections of such particles and it would be impossible to describe all the interactions between them. Therefore, the individual particles and their interactions are not taken into account: fluids are considered as a continuum. Essentially, fluid mechanics is the application of Newton's 2nd law of motion to a flowing continuum. Newton's 2nd law succinctly states:

$$F = m \frac{dv}{dt} \quad (1.1)$$

Or, the time rate of change in velocity (v) of a body is proportional to the net force (F) applied to that body. As a result, the change in the velocity of a small (infinitesimal) parcel of fluid is given by the sum of all the forces that operate on this fluid parcel. The picture on top of this page shows the example of blowing wind, but the same principles apply to ocean flows. All the time, various forces act on the flow; how the flow will develop, depends on all the forces taken together. For flows in the atmosphere and ocean, the collection of forces boils down to:

- Acceleration = [pressure gradient force](#) + [Coriolis force](#) + friction (horizontal)
- Acceleration = [pressure gradient force](#) + gravity + friction (vertical)

Or in mathematical terms:

$$\frac{du}{dt} = -\frac{\left(\frac{dp}{dx}\right)}{\rho} + f * v + K_h \left(\frac{d^2u}{dx^2} + \frac{d^2u}{dy^2}\right) + K_v \frac{d^2u}{dz^2} \quad (1.2)$$

$$\frac{dv}{dt} = -\frac{\left(\frac{dp}{dy}\right)}{\rho} - f * u + K_h \left(\frac{d^2v}{dx^2} + \frac{d^2v}{dy^2}\right) + K_v \frac{d^2v}{dz^2} \quad (1.3)$$

$$\frac{dw}{dt} = -\frac{\left(\frac{dp}{dz}\right)}{\rho} - g + K_h \left(\frac{d^2w}{dx^2} + \frac{d^2w}{dy^2}\right) + K_v \frac{d^2w}{dz^2} \quad (1.4)$$

Here, u is the velocity in the zonal (West to East) direction, v is the velocity in the meridional (South to North) direction, and w is the velocity in the vertical direction. Furthermore, p is the pressure, ρ is the density of the fluid, f the [Coriolis](#) parameter, and g is the gravitational acceleration. K_h and K_v are the horizontal and vertical friction coefficients; these are different, because on the large atmospheric and oceanic scales, momentum dispersion is primarily due to [turbulence](#), rather than molecular diffusion. In the following sections, each of these different forces will be discussed in more detail.

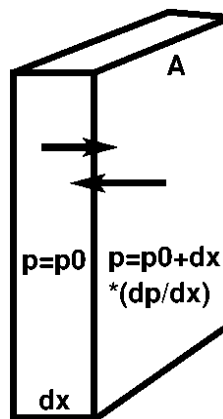
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2: Pressure Gradients

From high school geography class, you may remember that "air tends to flow from high pressure to low pressure". To understand why this happens, it is key to realize that gases (but also liquids) exert a force on their surroundings because of the thermal motion of the molecules. This force is in all directions, as the thermal motion is in all directions. The pressure (p) is simply the force (F) per unit area (A):

$$p = \frac{F}{A} \quad (2.1)$$

For gases, there exists a very simple approximate relationship between the pressure, the density, and the temperature of the gas: the Ideal Gas Law. Although no such an elegant formula exists for liquids, the International Equation of State is used to relate the pressure, temperature, salinity (salt concentration), and density of sea water. If, for whatever cause (temperature, density, salinity), the pressure at one point in space is different than at a neighboring point, then there exists a pressure gradient. Now, consider a parcel of fluid with infinitesimal thickness dx and surface area A as depicted in the Figure below, where there is a horizontal pressure gradient (dp/dx), with a higher pressure on the right than on the left.



The fluid on the right hand side of the parcel pushes the parcel to the left, whereas the fluid on the left hand side of the parcel pushes it to the right. Due to the pressure gradient, however, the push to the left is slightly harder than the push to the right. Thus, there is a net force on the parcel:

$$F = A * \left[p_0 - \left(p_0 + dx * \left(\frac{dp}{dx} \right) \right) \right] = -A * dx * \left(\frac{dp}{dx} \right) \quad (2.2)$$

According to [Newton's 2nd Law](#), the acceleration is:

$$\frac{dv}{dt} = \frac{F}{m} \quad (2.3)$$

and with $m = \rho * A * dx$, we finally obtain:

$$\frac{dv}{dt} = -\frac{\left(\frac{dp}{dx} \right)}{\rho} \quad (2.4)$$

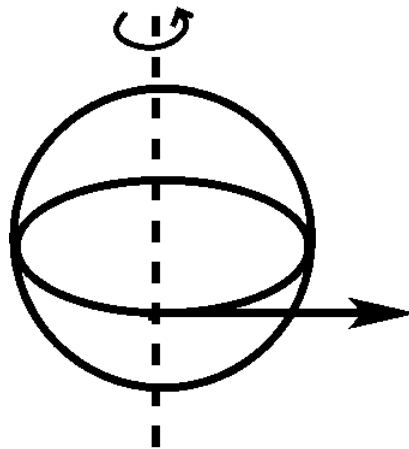
The minus sign indicates that the acceleration is in the opposite direction of the pressure gradient, that is, "air tends to flow from high pressure to low pressure"; the magnitude of the acceleration is equal to the pressure gradient divided by the density of the fluid.

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3: The Coriolis Force

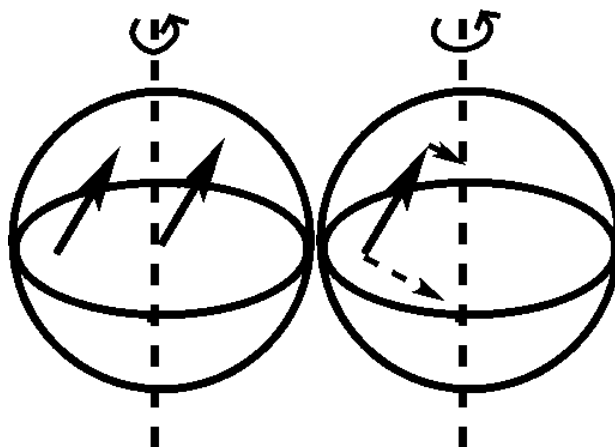
The Coriolis force is a central element of the dynamics of ocean and atmosphere flows. A rigorous derivation using vector calculus is given in many textbooks on classical mechanics, but such derivations are often not the most effective way to obtain a conceptual understanding. Therefore, we will provide a more intuitive qualitative explanation here. Throughout this chapter about [Geophysical Flows](#), we have been talking a lot about [Newton's 2nd law](#), but the key to understanding the Coriolis force is rather Newton's 1st law: an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force. The Earth and everything on it is constantly moving, as the Earth rotates around its axis. Moreover, the direction of all this movement changes constantly: while you are typing at your desk at 2 p.m., you are moving in the exact opposite direction compared to when you were lying in your bed at 2 a.m. These forced changes in the direction of the movement of objects that tend to keep moving in the same direction is what gives rise to so-called pseudo-forces (which are actually very real, if you happen to live on a rotating planet such as the Earth).

The pseudo-force that is the easiest to understand conceptually is the centrifugal force, depicted below.



Essentially, objects tend to be thrown off a rotating body, because the objects keep moving in a straight line (as indicated by the arrow), while the surface of the rotating body does not move in a straight line (as indicated by the circle). Fortunately, we don't need to worry about being thrown off the Earth, because the gravitational force that keeps our feet on the ground is about 300 times stronger than the centrifugal force. The centrifugal force plays no significant role in the dynamics of the ocean and the atmosphere, but it does make the Earth slightly elliptical.

The Coriolis force is a pseudo-force that emerges when an object is moving on the surface of a rotating body. The Coriolis force is conceptually much more subtle than the centrifugal force and, in fact, consists of two different effects, each illustrated below:



The left panel illustrates the first effect: an object that keeps moving in the same direction in absolute space appears to change direction, as the Earth turns to the right. At first, the object moves straight North toward the pole (left arrow), but then, the direction

of the movement becomes Northeast in the Earth frame of reference (right arrow), even though the direction and speed of the object have not changed in absolute space. The right panel illustrates the second effect: as an object moves on the Earth's surface (as indicated by the solid arrow), it carries the velocity of the Earth rotation (dashed arrows) with it. As the object moves North, it comes into regions where the velocity of the Earth rotation is smaller than at its original location. Thus, it will have an excess velocity to the East which deflects the movement from the North toward the Northeast, when viewed in the Earth frame of reference. In the Earth frame of reference, each of these 2 effects leads to an acceleration with a magnitude equal to $\Omega * v \sin \theta$ (with Ω the angular velocity of the Earth, v the velocity of the object in the Earth frame of reference, and θ the latitude), so the total acceleration adds up to $2 * \Omega * v \sin \theta$ or $f * v$, with the Coriolis parameter $f = 2 * \Omega * \sin \theta$. The Coriolis force is at a 90° angle with respect to the movement of the object, to the right in the Northern hemisphere, to the left in the Southern Hemisphere; it vanishes at the Equator ($\theta = 0$). In the zonal and meridional directions, the accelerations amount to:

$$\frac{du}{dt} = f * v \quad (3.1)$$

$$\frac{dv}{dt} = -f * u \quad (3.2)$$

The impact of the Coriolis force on moving parcels of air (or water) is [demonstrated here](#).

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4: Turbulent Diffusion

On the scale of the ocean, diffusion due to the random thermal motion of molecules is very slow. With a diffusion coefficient of $\sim 2 \times 10^{-9} \text{ m}^2/\text{s}$, diffusive transport over a distance of just 100 m takes about 100000 years. However, transport rates in both the ocean and the atmosphere are enhanced by many orders of magnitude through stirring by eddies of various sizes that are collectively named 'turbulence'. The study of turbulence forms a very large and complicated topic that we are not going to cover in-depth here. Instead, we will try to provide an idea of turbulence as a phenomenon and of the role it plays in the ocean and the atmosphere.

Turbulence can have a mechanical or a convective origin. As an everyday example of mechanical production of turbulence, take a look at this instructional video by Tony Kuphaldt of Bellingham Technical College.



As the tap is opened further and further and the velocity of the water increases, the flow changes from laminar (regular and transparent) to turbulent which means that it is full of little eddies. At the ocean surface, vigorous mechanical turbulence production can take place when a storm blows over the water. Turbulence is produced convectively, when a fluid is heated from below or cooled from above. The cooler, denser fluid on top tends to sink, while the warmer, lighter fluid at the bottom tends to rise, leading to the formation of eddies. When the ocean surface cools during winter (particularly at high latitudes), the result is convective turbulence production. However, in most places, the ocean is stably stratified most of the time, that is, the ocean surface is warmer than the deeper water. This tends to dampen true three-dimensional turbulence which is why oceanic turbulence is to a large extent two-dimensional. Therefore, the horizontal turbulent diffusivity (K_h) is typically orders of magnitude larger than the vertical turbulent diffusivity (K_v). Turbulence, whether two- or three-dimensional, can lead to rapid and erratic dispersion. Overall, the effective dispersion of the momentum of the large-scale background flow over rather large distances is the main reason why turbulence is important for the large-scale ocean circulation. To a satisfactory approximation, this can be represented as a diffusion process with different horizontal and vertical diffusion coefficients:

$$\frac{dv}{dt} = K_h \left(\frac{d^2v}{dx^2} + \frac{d^2v}{dy^2} \right) + K_v \frac{d^2v}{dz^2} \quad (4.1)$$

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5: Geostrophic Balance

Now that we have an understanding of the main forces acting on fluid parcels moving through the ocean and the atmosphere, we focus on how these forces impact the flows in the atmosphere and the ocean. In the horizontal momentum balance equations (1.2a) and (1.2b) in Section 1, the pressure gradient and Coriolis terms are almost always much larger than the other terms. Therefore, to a good approximation, there exists a balance between the **pressure gradient** and **Coriolis forces** named **geostrophic balance**:

$$\frac{\left(\frac{dp}{dx}\right)}{\rho} = f * v \quad (5.1)$$

$$\frac{\left(\frac{dp}{dy}\right)}{\rho} = -f * u \quad (5.2)$$

Geostrophic balance is arguably the most central concept in physical oceanography and dynamical meteorology. Although higher-order processes are responsible for all the interesting dynamics, it is important to realize that almost all large-scale flows in the atmosphere and ocean are in geostrophic balance to leading order. The only place where geostrophic balance never holds is at the Equator where there is no Coriolis force. A key feature of geostrophic balance is that rather than flowing from high to low pressure, the fluid actually moves parallel to lines of equal pressure (isobars). Viewing in the direction of the flow, low pressure is to the left and high pressure is to the right on the Northern Hemisphere (and vice versa in the Southern Hemisphere). This is illustrated in Figure 5.1.

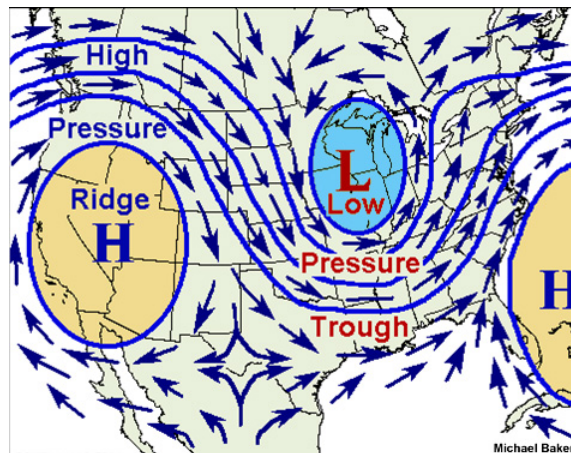


Figure 5.1: the weather map (National Weather Service, Southern Region Headquarters).

To understand conceptually how geostrophic balance emerges, imagine a fluid parcel that starts accelerating due to a pressure gradient. As the parcel moves, it is deflected by the **Coriolis force** (see left panel below), until it flows parallel to the isobars and the pressure gradient and Coriolis force cancel each other (right panel below).

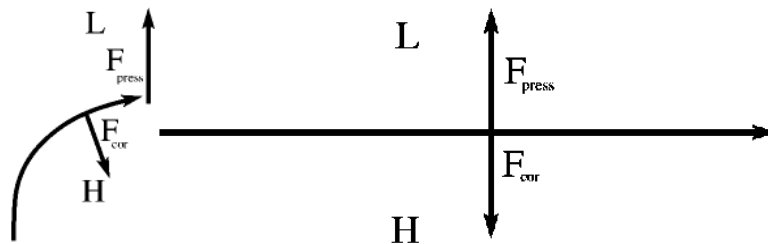


Figure 5.2: Paste Caption Here

From the Equations 5.1 and 5.2, it can easily be seen that

$$u_g = -\frac{\left(\frac{dp}{dy}\right)}{f * \rho} \quad (5.3)$$

and

$$v_g = \frac{\left(\frac{dp}{dx}\right)}{f * \rho}, \quad (5.4)$$

with u_g and v_g the geostrophic velocities in the zonal and meridional directions, respectively. Thus, the geostrophic velocities increase with an increasing pressure gradient. Somewhat counterintuitively, geostrophic velocities decrease with increasing latitude, although the Coriolis force is stronger at higher latitudes. This is because with a weaker Coriolis force, the fluid is able to accelerate more under the influence of the pressure gradient force, before geostrophic equilibrium is reached.

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6: Wind Stress

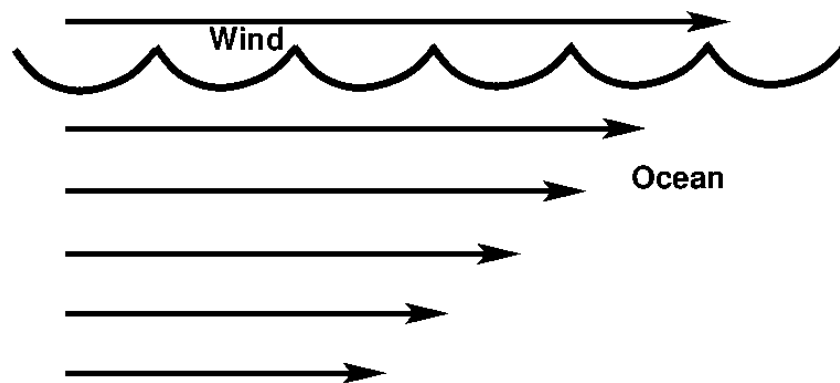
The large-scale currents at the ocean surface, such as the Gulf Stream from Florida to Europe, the Antarctic Circumpolar Current, and the Kuroshio from the Philippines to Japan, are all driven by the wind. In fact, it has been estimated that the winds do ~ 1 TW worth of mechanical work on the oceans. How do the winds do this? As the wind blows over the ocean, it exerts a force on the ocean surface in the direction of the wind. This force, per unit ocean surface area, is what is called the wind stress (τ_w). At the same time, the ocean water resists this wind stress through internal friction. According to Newton's law of viscosity, this frictional force per unit area is proportional to the velocity gradient in the direction perpendicular to the surface:

$$\tau_v = \rho K_v \frac{dv}{dz}. \quad (6.1)$$

Again, we use the **turbulent diffusion** coefficient K_v , rather than a molecular viscosity. In steady state, the wind stress and the frictional force must cancel, that is: $\tau_w = \tau_v$. Therefore, we have:

$$\tau_w = \rho K_v \frac{dv}{dz} \quad (6.2)$$

Overall, the force exerted by the wind on the ocean surface gives rise to a vertical velocity gradient in the upper boundary layer of the ocean, as depicted below. Essentially, the wind adds momentum to the ocean surface which is then transported downward through diffusion associated with the vertical momentum gradient.



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7: Basis of Wind-Driven Circulation - Ekman spiral and transports

In [Section 6](#), it was mentioned that the large-scale currents at the ocean surface are all driven by the wind. This seems logical enough at first sight, but the Arctic explorer Fridtjof Nansen noticed something strange: icebergs tend to drift at an angle to the right of the prevailing wind direction. To explain this remarkable observation, [Ekman \(1905\)](#) formulated a theory that is still a cornerstone of physical oceanography. The central assumption is that near the ocean surface, the largest deviations from [geostrophic balance](#) occur as a result of the [wind stress](#) which leads to [momentum diffusion](#) in the vertical direction. This means that to a good approximation, the horizontal momentum balance equations (1.2a) and (1.2b) in [Section 1](#) become:

$$\frac{\left(\frac{dp}{dx}\right)}{\rho} = f * v + K_v \frac{d^2 u}{dz^2} \quad (7.1)$$

$$\frac{\left(\frac{dp}{dy}\right)}{\rho} = -f * u + K_v \frac{d^2 v}{dz^2} \quad (7.2)$$

We now split the velocity up in a [geostrophic](#) part (u_g, v_g) and an ageostrophic Ekman velocity (u_E, v_E) :

$$\frac{\left(\frac{dp}{dx}\right)}{\rho} = f * (v_g + v_E) + K_v \frac{d^2 (u_g + u_E)}{dz^2} \quad (7.3)$$

$$\frac{\left(\frac{dp}{dy}\right)}{\rho} = -f * (u_g + u_E) + K_v \frac{d^2 (v_g + v_E)}{dz^2} \quad (7.4)$$

From equations (5.1a) and (5.1b) in [Section 5](#), we can see that the geostrophic velocities cancel against the pressure gradient terms on the lefthand side; the terms $K_v \frac{d^2 u_g}{dz^2}$ and $K_v \frac{d^2 v_g}{dz^2}$ can be neglected. Therefore, the equations simplify to:

$$f * v_E = -K_v \frac{d^2 u_E}{dz^2} \quad (7.5)$$

$$f * u_E = K_v \frac{d^2 v_E}{dz^2} \quad (7.6)$$

which can be reformulated through substitution into one fourth-order ordinary differential equation:

$$u_E = -\left(\frac{K_v}{f}\right)^2 \frac{d^4 u_E}{dz^4} \quad (7.7)$$

with the (real) solution:

$$u_E = A_1 \cos\left(\sqrt{\frac{f}{2K_v}} z + \phi_1\right) e^{\sqrt{\frac{f}{2K_v}} z} + A_2 \cos\left(\sqrt{\frac{f}{2K_v}} z + \phi_2\right) e^{-\sqrt{\frac{f}{2K_v}} z} \quad (7.8)$$

To determine the different coefficients, we use two boundary conditions:

1. The direct impact of the [wind stress](#) disappears in the deep ocean: $u_E \rightarrow 0$ for $z \rightarrow -\infty$. Therefore, A_2 must be equal to 0.
2. In [Section 6](#), we argued that close to the ocean-atmosphere interface, the [wind stress](#) is linearly proportional to the vertical velocity gradient; this means that if the wind is blowing in the zonal (West-East) direction, $\frac{du_E}{dz} = \frac{\tau_w}{\rho K_v}$ (equation 6.1), $\frac{dv_E}{dz} = 0$ for $z = 0$. This leads to $\phi_1 = -\frac{\pi}{4}$, $A_1 = \frac{\tau_w}{\rho \sqrt{f K_v}}$.

Overall, we have:

$$u_E = \frac{\tau_w}{\rho \sqrt{f K_v}} \cos\left(\sqrt{\frac{f}{2K_v}} z - \frac{\pi}{4}\right) e^{\sqrt{\frac{f}{2K_v}} z} \quad (7.9)$$

$$v_E = \frac{\tau_w}{\rho \sqrt{f K_v}} \sin\left(\sqrt{\frac{f}{2K_v}} z - \frac{\pi}{4}\right) e^{\sqrt{\frac{f}{2K_v}} z}$$

$$v_E = -\frac{\rho_v}{f} \frac{du_E}{dz} = \frac{\tau_w}{\rho \sqrt{f K_v}} \sin \left(\sqrt{\frac{f}{2 K_v}} z - \frac{\pi}{4} \right) e^{-\sqrt{2 K_v} z} \quad (7.10)$$

The Ekman transports per unit area in the zonal and meridional directions are respectively:

$$M_{E,x} = \rho \int_{-\infty}^0 u_E dz \quad (7.11)$$

$$M_{E,y} = \rho \int_{-\infty}^0 v_E dz \quad (7.12)$$

These could be calculated by integrating Equations 7.9 and 7.10, but it is much easier to use Equations 7.5 and 7.6:

$$M_{E,x} = \frac{K_v \rho}{f} \int_{-\infty}^0 \frac{d^2 v_E}{dz^2} dz \quad (7.13)$$

$$= \frac{K_v \rho}{f} \left(\frac{dv_E}{dz}(z=0) - \frac{dv_E}{dz}(z \rightarrow -\infty) \right) \quad (7.14)$$

$$= 0 \quad (7.15)$$

and

$$M_{E,y} = -\frac{K_v \rho}{f} \int_{-\infty}^0 \frac{d^2 u_E}{dz^2} dz \quad (7.16)$$

$$= -\frac{K_v \rho}{f} \left(\frac{du_E}{dz}(z=0) - \frac{du_E}{dz}(z \rightarrow -\infty) \right) \quad (7.17)$$

$$= -\frac{\tau_w}{f} \quad (7.18)$$

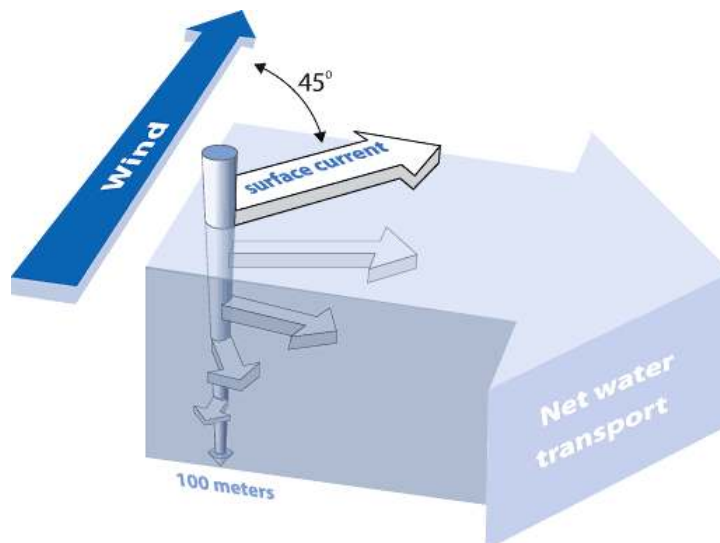
What does all this mean? At the ocean surface,

$$u_E = \frac{\tau_w}{\rho \sqrt{2 f K_v}}$$

and

$$v_E = -\frac{\tau_w}{\rho \sqrt{2 f K_v}}$$

via Equations 7.9 and 7.10, that is, the Ekman velocity is at an angle of 45° to the right of the wind direction in the Northern Hemisphere (and to the left of the wind in the Southern Hemisphere) due to the Coriolis force. Going deeper, the Coriolis force keeps turning the direction of the flow further to the right, while the water speed decreases exponentially with depth. As illustrated in the Figure below (courtesy of NOAA), the overall flow pattern forms a so-called Ekman spiral. Furthermore, Equation 7.15 and 7.18 imply that the net Ekman transport is at 90° to the right of the wind direction in the Northern Hemisphere.



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9: Vorticity

Before the discussion of ocean currents, there is one more key concept that needs to be introduced: vorticity. Essentially, vorticity is the same as rotation in a horizontal plane. Mathematically, the vorticity ζ is defined as:

$$\zeta = \frac{dv}{dx} - \frac{du}{dy} \quad (9.1)$$

Which is also just the [curl](#) of velocity.

$$\zeta = \vec{\nabla} \times \vec{v} \quad (9.2)$$

Positive vorticity means that the fluid rotates counterclockwise, whereas negative vorticity implies clockwise rotation. Using the horizontal momentum balance equations (1.2a) and (1.2b) from [Section 1](#), a vorticity equation can be constructed, describing the time development of the vorticity of a fluid parcel. Under the assumption that the density of the fluid is constant, this equation becomes:

$$\frac{d\zeta}{dt} = \frac{d}{dx} \left(\frac{dv}{dt} \right) - \frac{d}{dy} \left(\frac{du}{dt} \right) = -f \left(\frac{du}{dx} + \frac{dv}{dy} \right) - \beta v + K_h \left(\frac{d^2\zeta}{dx^2} + \frac{d^2\zeta}{dy^2} \right) + K_v \frac{d^2\zeta}{dz^2} \quad (9.3)$$

with $\beta = \frac{df}{dy}$. The terms on the right-hand side of the equation can be interpreted as follows: $-f \left(\frac{du}{dx} + \frac{dv}{dy} \right)$ says that horizontal divergence or convergence of the flow leads to rotation due to the [Coriolis force](#); βv is the so-called β -effect: as a parcel moves in the meridional (North-South) direction, it tends to spin up, because the Coriolis force is stronger on one side of the parcel than on the other side; the further terms simply indicate turbulent diffusion of vorticity.

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T

turbulence

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vertical turbulent diffusivity

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vorticity

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